

Lecture 4 Exercise Solutions

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1 Important Equations

Joint Probability Mass Function

The joint probability mass function (joint pmf) of the discrete random variable X and Y , denoted as $f_{XY}(x, y)$, satisfies

1. $f_{XY}(x, y) \geq 0$
2. $\sum_X \sum_Y f_{XY}(x, y) = 1$
3. $f_{XY}(x, y) = P(X = x, Y = y)$

Marginal pmf

The discrete marginal pmf is defined as:

$$f_X(x) = \sum_y f_{XY}(x, y) \quad (1)$$

Moments

The mean of X can be calculated as:

$$\mu_X = E[X] = \sum x f_X(x) \quad (2)$$

And the variance as:

$$\sigma_X^2 = V[X] = E[X^2] - \mu_X^2 \quad (3)$$

This can of course also be done for Y in the same way.

Joint Probability Density Function

A joint probability density function for the continuous random variables X and Y , denoted as $f_{XY}(x, y)$, satisfies the following properties:

1. $f_{XY}(x, y) \geq 0$ for all x, y
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$
3. For any region R of two-dimensional space,

$$P((X, Y) \in R) = \int \int_R f_{XY}(x, y) dx dy \quad (4)$$

Covariance

The covariance between two variables X and Y is defined as:

$$\text{cov}(X, Y) = \sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] \quad (5)$$

Correlation Coefficient

The correlation coefficient measures the amount of linear correlation between X and Y :

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad (6)$$

Where $-1 \leq \rho_{XY} \leq 1$.

Unilinear function of X

For an unilinear function of X on the form $Y = h(x)$, with a given pdf for X we can find the pdf for Y as:

$$f_Y(y) = f_X[h^{-1}(y)] \left| \frac{dh^{-1}(y)}{dy} \right| \quad (7)$$

2 Exercise 5.1.2

Show that the following function satisfies the properties of a joint probability mass function.

x	y	$f_{XY}(x, y)$
-1.0	-2	$\frac{1}{8}$
-0.5	-1	$\frac{1}{4}$
0.5	1	$\frac{1}{2}$
1.0	2	$\frac{1}{8}$

First we check that this satisfies the criteria defined above. We see that 1. is satisfied as $0 \leq f_{XY}(x, y) \leq 1$. We can now check 2. by summing:

$$\sum_X \sum_Y f_{XY}(x, y) = \frac{1}{8} + \frac{1}{4} + \frac{1}{2} + \frac{1}{8} = 1$$

Meaning the function satisfies the properties of a joint probability mass function.

Determine the following:

a. $P(X < 0.5, Y < 1.5)$

To find the probability we look at the table and sum the values for which $X < 0.5$

and $Y < 1.5$:

$$P(X < 0.5, Y < 1.5) = f_{XY}(-1, -2) + f_{XY}(-0.5, -1) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

b. $P(X < 0.5)$

We now only have a bound for X however this is the same as in a. meaning:

$$P(X < 0.5) = \frac{3}{8}$$

c. $P(Y < 1.5)$

Now there is only a bound on Y meaning we sum all values of the function below this limit:

$$P(Y < 1.5) = f_{XY}(-1, -2) + f_{XY}(-0.5, -1) + f_{XY}(0.5, 1) = \frac{1}{8} + \frac{1}{4} + \frac{1}{2} = \frac{7}{8}$$

d. $P(X > 0.25Y < 4.5)$

Here we see that two values satisfy this condition:

$$P(X > 0.25Y < 4.5) = f_{XY}(0.5, 1) + f_{XY}(1, 2) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}$$

e. $E(X), E(Y), V(X), V(Y)$

First we calculate the mean of X and Y using eq. 2:

$$\begin{aligned}\mu_X &= (-1) \cdot \frac{1}{8} + (-0.5) \cdot \frac{1}{4} + 0.5 \cdot \frac{1}{2} + 1 \cdot \frac{1}{8} = \frac{1}{8} \\ \mu_Y &= (-2) \cdot \frac{1}{8} + (-1) \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{8} = \frac{1}{4}\end{aligned}$$

And the variance using eq. 3:

$$\begin{aligned}\sigma_X^2 &= (-1)^2 \cdot \frac{1}{8} + (-0.5)^2 \cdot \frac{1}{4} + 0.5^2 \cdot \frac{1}{2} + 1^2 \cdot \frac{1}{8} - \left(\frac{1}{8}\right)^2 = 0.422 \\ \sigma_Y^2 &= (-2)^2 \cdot \frac{1}{8} + (-1)^2 \cdot \frac{1}{4} + 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{8} - \left(\frac{1}{4}\right)^2 = 1.69\end{aligned}$$

f. Marginal probability distribution of X

We can find the marginal probability distribution of X by ignoring the Y values which gives:

x	$f_{XY}(x, y)$
-1.0	$\frac{1}{8}$
-0.5	$\frac{1}{4}$
0.5	$\frac{1}{2}$
1.0	$\frac{1}{8}$

3 Exercise 5.1.9

Determine the value of c such that the function $f(x, y) = cxy$ for $0 < x < 3$ and $0 < y < 3$ satisfies the properties of a joint probability density function.

Determine the following:

To find the value of c we normalize the function inside the given domain using eq. 4 since the probability is zero outside:

$$\begin{aligned} 1 &= \int_0^3 \int_0^3 cxy \, dx \, dy = c \int_0^3 \int_0^3 xy \, dx \, dy = c \int_0^3 \left. \frac{1}{2}x^2 \right|_0^3 y \, dy = 3c \int_0^3 y \, dy \\ &= \frac{9}{2}c \int_0^3 y \, dy = \frac{9}{2}c \left[\frac{1}{2}y^2 \right]_0^3 = \frac{9}{2}c \left(\frac{9}{2} \right) = \frac{81}{4}c \Rightarrow c = \frac{4}{81} \end{aligned}$$

Meaning the function is:

$$f_{XY}(x, y) = \frac{4}{81}xy$$

a. $P(X < 2, Y < 3)$

We again use eq. 4 and integrate in the given domain:

$$\begin{aligned} P(X < 2, Y < 3) &= \frac{4}{81} \int_0^3 \int_0^2 xy \, dx \, dy = \frac{4}{81} \int_0^3 \left. \frac{1}{2}x^2 \right|_0^2 y \, dy = \frac{8}{81} \int_0^3 y \, dy = \frac{8}{81} \left[\frac{1}{2}y^2 \right]_0^3 \\ &= \frac{8}{81} \left(\frac{9}{2} \right) = \frac{4}{9} \end{aligned}$$

b. $P(X < 2.5)$

Here we have no bound on Y so we integrate in its the whole domain while we have

a limit on X :

$$\begin{aligned} P(X < 2.5) &= \frac{4}{81} \int_0^3 \int_0^{2.5} xy \, dx dy = \frac{4}{81} \int_0^3 \left. \frac{1}{2} x^2 \right|_0^{2.5} y \, dy = 0.154 \int_0^3 y \, dy = 0.154 \left[\frac{1}{2} y^2 \right]_0^3 \\ &= 0.154 \left(\frac{9}{2} \right) = 0.694 \end{aligned}$$

c. $P(1 < Y < 2.5)$

We integrate in the whole domain of X and have limited bounds on Y :

$$\begin{aligned} P(1 < Y < 2.5) &= \frac{4}{81} \int_1^{2.5} \int_0^3 xy \, dx dy = \frac{4}{81} \int_1^{2.5} \left. \frac{1}{2} x^2 \right|_0^3 y \, dy = \frac{2}{9} \int_1^{2.5} y \, dy = \frac{2}{9} \left[\frac{1}{2} y^2 \right]_1^{2.5} \\ &= \frac{1}{9} (2.5^2 - 1^2) = 0.583 \end{aligned}$$

d. $P(X > 1.8, 1 < Y < 2.5)$

Inserting the given bounds in the integral limits:

$$\begin{aligned} P(X > 1.8, 1 < Y < 2.5) &= \frac{4}{81} \int_1^{2.5} \int_{1.8}^3 xy \, dx dy = \frac{4}{81} \int_1^{2.5} \left. \frac{1}{2} x^2 \right|_{1.8}^3 y \, dy \\ &= \frac{2}{81} \int_1^{2.5} (3^2 - 1.8^2) y \, dy = 0.142 \int_1^{2.5} y \, dy \\ &= 0.142 \left[\frac{1}{2} y^2 \right]_1^{2.5} = 0.071 (2.5^2 - 1^2) = 0.373 \end{aligned}$$

e. $E[X]$

We find the mean of X as:

$$\begin{aligned} E[X] &= \int_0^3 \int_0^3 x f_{XY}(x, y) \, dx dy = \frac{4}{81} \int_0^3 \int_0^3 x^2 y \, dx dy = \frac{4}{81} \int_0^3 \left. \frac{1}{3} x^3 \right|_0^3 y \, dy \\ &= \frac{4}{9} \int_0^3 y \, dy = \frac{4}{9} \left[\frac{1}{2} y^2 \right]_0^3 = \frac{2}{9} (9) = 2 \end{aligned}$$

f. $P(X < 0, Y < 4)$

Since X is only defined in the domain $[0; 3]$ then this probability is $P(X < 0, Y < 4) = 0$.

g. Marginal probability distribution of X

We find the marginal probability distribution of X by integrating over Y :

$$f_X(x) = \int_a^b f_{XY}(x, y) \, dy = \frac{4}{81} \int_0^3 xy \, dy = \frac{4}{81} x \left[\frac{1}{2} y^2 \right]_0^3 = \frac{2}{81} x (9) = \frac{2}{9} x$$

4 Exercise 5.4.1

Determine the covariance and correlation for the following joint probability distribution:

x	1	1	2	4
y	3	4	5	6
$f_{XY}(x, y)$	1/8	1/4	1/2	1/8

To find the covariance we first need to find the mean of the variables:

$$\mu_X = 1 \cdot \frac{1}{8} + 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{2} + 4 \cdot \frac{1}{8} = 1.875$$

$$\mu_Y = 3 \cdot \frac{1}{8} + 4 \cdot \frac{1}{4} + 5 \cdot \frac{1}{2} + 6 \cdot \frac{1}{8} = 4.63$$

The covariance can now be calculated using eq. 5:

$$\begin{aligned} \sigma_{XY} &= (1 - 1.875)(3 - 4.63)\frac{1}{8} + (1 - 1.875)(4 - 4.63)\frac{1}{4} + (2 - 1.875)(5 - 4.63)\frac{1}{2} \\ &\quad + (4 - 1.875)(6 - 4.63)\frac{1}{8} = 0.703 \end{aligned}$$

To calculate the correlation we now need to find the standard deviation of the variables first:

$$\begin{aligned} \sigma_X &= \sqrt{1^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{1}{4} + 2^2 \cdot \frac{1}{2} + 4^2 \cdot \frac{1}{8} - 1.875^2} = 0.927 \\ \sigma_Y &= \sqrt{3^2 \cdot \frac{1}{8} + 4^2 \cdot \frac{1}{4} + 5^2 \cdot \frac{1}{2} + 6^2 \cdot \frac{1}{8} - 4.63^2} = 0.857 \end{aligned}$$

The correlation coefficient is now calculated using eq. 6:

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} = \frac{0.703}{0.927 \cdot 0.857} = 0.885$$

5 Exercise 5.4.7

The joint probability distribution is

x	-1	0	0	1
y	0	-1	1	0
$f_{XY}(x, y)$	1/4	1/4	1/4	1/4

Show that the correlation between X and Y is zero but X and Y are not independent.

Looking at the table we can see that the mean of both X and Y is zero. This can also be proven by calculation:

$$\mu_X = (-1)\frac{1}{4} + 0\frac{1}{4} + 0\frac{1}{4} + 1\frac{1}{4} = 0, \quad \mu_Y = 0\frac{1}{4} + (-1)\frac{1}{4} + 1\frac{1}{4} + 0\frac{1}{4} = 0$$

Or the combined mean:

$$\mu_{XY} = \sum xyf_{XY}(x, y) = 0$$

The standard deviation is:

$$\sigma_X = \sqrt{(-1)^2\frac{1}{4} + 1^2\frac{1}{4} - 0^2} = 0.707, \quad \sigma_Y = \sqrt{(-1)^2\frac{1}{4} + 1^2\frac{1}{4} - 0^2} = 0.707$$

The covariance is calculated:

$$\sigma_{XY} = \mu_{XY} - \mu_X\mu_Y = 0 - 0 \cdot 0 = 0$$

Meaning the correlation coefficient is $\rho_{XY} = 0$ i.e. no correlation. However, X and Y are still dependent, since when one of them is 1 or -1 the other is 0.

6 Exercise 5.6.1

X and Y are independent normal variables with $E[X] = 2$, $V[X] = 5$, $E[Y] = 6$, and $V[Y] = 8$. Determine the following:

a. $E[3X + 2Y]$

For the mean we can use superposition principle and linearity meaning $E[aX] = aE[X]$:

$$E[3X + 2Y] = E[3X] + E[2Y] = 3E[X] + 2E[Y] = 3 \cdot 2 + 2 \cdot 6 = 18$$

b. $V[3X + 2Y]$

Because X and Y are independent, superposition holds also for variance. We also use that $V[aX] = a^2V[X]$:

$$V[3X + 2Y] = V[3X] + V[2Y] = 3^2V[X] + 2^2V[Y] = 9 \cdot 5 + 4 \cdot 8 = 77$$

c. $P(3X + 2Y < 18)$

The variables follow the normal distribution with $N(\mu, \sigma^2) = N(18, 77)$. We use standardization:

$$P(3X + 2Y < 18) \Rightarrow P\left(Z < \frac{X - \mu}{\sigma}\right) = P\left(Z < \frac{18 - 18}{\sqrt{77}}\right) = P(Z < 0)$$

We can solve this in MATLAB as:

$$P(Z < 0) = \text{normcdf}(0) = \frac{1}{2}$$

Meaning $P(3X + 2Y < 18) = 50\%$. This also makes intuitively sense since we are looking at the probability of a value below the mean following a normal distribution, which is symmetric.

d. $P(3X + 2Y < 28)$

We do the same as before:

$$P(3X + 2Y < 28) \Rightarrow P\left(Z < \frac{28 - 18}{\sqrt{77}}\right) = P(Z < 1.14)$$

Solving with MATLAB "normcdf(1.14)" returns:

$$P(Z < 1.14) = 0.873$$

7 Exercise 5.6.9

Consider the perimeter of a part in Example 5.27. Let X_1 and X_2 denote the length and width of a part with standard deviations 0.1 and 0.2 centimeters, respectively. Suppose that the covariance between X_1 and X_2 is 0.02. Determine the variance of the perimeter $Y = 2X_1 + 2X_2$ of a part. Compare and comment on the result here and in the example.

When we have a linear combination of dependent variables $Y = c_0 + c_1X_1 + c_2X_2 + \dots + C_pX_p$ we can find the variance as (eq. 5.27 in the book):

$$V[Y] = c_1^2V[X_1] + c_2^2V[X_2] + \dots + c_p^2V[X_p] + 2 \sum_{i < j} c_i c_j \text{cov}(X_i, X_j)$$

We insert the know values with $Y = 2X_1 + 2X_2$:

$$V[Y] = 2^2 \cdot 0.1^2 + 2^2 \cdot 0.2^2 + 2 \cdot 2 \cdot 2 \cdot 0.02 = 0.36$$

We see that there is a higher variance for the perimeter than in the example due to the covariance since they are no longer independent.

8 Exercise 5.7.3

A random variable X has the probability distribution

$$f_X(x) = e^{-x}, \quad x \geq 0$$

Determine the probability distribution for the following:

a. $Y = X^2$

We first find the pdf of Y using eq. 7:

$$y = h(x) = x^2 \Rightarrow x = \sqrt{y} = h^{-1}(y)$$

We insert:

$$f_Y(y) = f_X(\sqrt{y}) \frac{d}{dy} \sqrt{y} = f_X(\sqrt{y}) \frac{1}{2\sqrt{y}} = \frac{e^{-\sqrt{y}}}{2\sqrt{y}}$$

b. $Y = \sqrt{X}$

We use the same strategy as before:

$$y = \sqrt{x} \Rightarrow x = y^2 = h^{-1}(y)$$

Inserting:

$$f_Y(y) = f_X(y^2) \frac{d}{dy} y^2 = f_X(y^2) 2y = 2ye^{-y^2}$$

c. $Y = \ln(X)$

$$y = \ln(x) \Rightarrow x = e^y \rightarrow f_Y(y) = f_X(e^y) \frac{d}{dy} e^y = e^{-e^y} e^y = e^{y-e^y}$$